

Particles as Localized Persistence Classes

The Minimal Object Role of Particle Identity under Field-Carrier Mediation

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Abstract

The present work derives the minimal persistence-class role of particle identity within the realized-physics arc. Previous results established that realized persistence cannot be physically null, that standing-bearing persistence localizes, that localized standing induces continuation-neighborhood domains, that stable comparison supplies the metric role of a quotient comparison carrier, that perturbation transport is bounded by continuation-comparison constraints, and that localized persistence under such transport requires field-carrier mediation. It asks what makes a particle identifiable in that regime. It shows that primitive objecthood, point occupancy, trajectory labels, field labels, hidden cores, measurement records, and bookkeeping names are insufficient to ground particle identity. A particle is the stable localized persistence class of a carrier-mediated configuration.

The main result is a particle persistence-class theorem. In a fixed admissible realized persistence regime with field-carrier assignment Φ_D , bounded perturbation transport, local continuation domains, and quotient comparison carrier C_D , any localized carrier-mediated configuration whose standing signature and boundary/classification witness remain stable under admissible continuation induces a particle-role class

$$\Pi_D = [\kappa_U]_{\equiv \Pi}.$$

The equivalence relation $\equiv_{\Pi}$ preserves standing signature, carrier mediation role, boundary/classification witness, quotient comparison, restoration profile, identity-memory, local-domain role, and admitted transport role. This is the minimal physical persistence-class role later specialized by particle species, quantum states, excitations, Standard Model classifications, and detection records. No point-particle ontology, worldline primitive, Fock-space formalism, spin-statistics theorem, gauge representation, field equation, scattering theory, measurement-outcome ontology, or hidden-variable model is assumed.

1 Introduction

The realized-physics arc has so far derived increasingly recognizable physical roles without importing the later mathematical form of those roles as primitives. The first paper establishes the non-null floor of physical realization: realized persistence cannot be reconstruction-null inside its own admissible continuation regime [1]. The second paper derives bounded standing-localization as the origin of physical locality [2]. The third derives continuation-neighborhood domains as the origin of local-domain structure [3]. The fourth derives the metric role as quotient-stable comparative representation among local continuation domains [4]. The fifth derives bounded perturbation transport as the finite-propagation constraint role [5]. The sixth derives field-carrier mediation as the minimal carrier role of physical field structure [6].

In compact form, the chain is

$$\text{RealPers}_D \Rightarrow \text{NonVacRec}_D \Rightarrow [B_D]_{\equiv_B} \Rightarrow (\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D) \Rightarrow C_D \Rightarrow \text{BoundTrans}_D \Rightarrow \Phi_D.$$

The present analysis begins at the final term. Localized persistence is now supported by a distributed field-carrier assignment Φ_D . The next physical question is not optional:

What makes a localized carrier-mediated configuration identifiable as the same particle across continuation, transport, perturbation, restoration, and boundary comparison?

Primitive objecthood, point occupancy, trajectory labels, field labels, hidden cores, and measurement records are insufficient to ground that identity by themselves. Each can become identity-relevant only when it preserves standing-bearing class data under admissible continuation; otherwise it either records a description, imports a later specialization as primitive, or fails to preserve identity. Particle identity first enters realized physics as stable localized persistence-class identity within the field-carrier regime.

1.1 Theorem claim and physical role

The theorem-level claim is

$$(\Phi_D, \text{BoundTrans}_D, C_D, \text{LCD}_D) \wedge \text{StableLoc}_D \wedge \text{IdRet}_D \Rightarrow \text{ParticleRole}_D.$$

The physical-role claim is that ParticleRole_D is the minimal persistence-class role occupied by particle descriptions: a stable equivalence class of localized carrier-mediated configurations. The result is prior to the standard formal species of particle physics. It does not assume point particles, worldlines, quantum excitations, Fock states, particle-number operators, scattering amplitudes, mass eigenstates, gauge representations, or Standard Model species. Rather, it derives the structural role that those later formalisms must occupy if they are to describe particles as identifiable physical entities.

This distinction matters. The paper does not deny the usefulness of particle object descriptions. It shows that primitive objecthood alone is insufficient to ground particle identity. On the present reconstruction, objecthood is the stabilized appearance of a localized persistence class: the persistence-class role is what later object descriptions track.

Paper	Structural result	Physical role
1	NonVacRec_D	physical realization cannot be null
2	$[B_D]_{\equiv_B}$	origin of physical locality
3	$(\text{Adj}_D, \mathcal{N}_D, \text{LCD}_D)$	origin of local-domain organization
4	$C_D = \text{CompWit}_D / \sim_D$	origin of metric comparison role
5	BoundTrans_D	origin of finite-propagation constraint
6	Φ_D	origin of field-carrier mediation
7	$\Pi_D = [\kappa_U]_{\equiv_\Pi}$	origin of particle identity

1.2 Scope

The paper derives the minimal particle role. It does not derive electron, muon, quark, neutrino, gauge-boson, or Higgs species. It does not derive spin-statistics, quantum measurement theory, Fock-space excitations, cross sections, mass values, Standard Model representations, worldline

dynamics, or creation and annihilation operators. Those are later specializations. What is derived here is the prior identity role: a carrier-mediated localized configuration is particle-identifiable only when its standing signature and boundary/classification witness remain stable under the admissible operations of the realized regime.

2 Dependencies and imported structure

The paper uses the following prior structure.

Imported result	Role in the present paper
Non-vacuous realized persistence	A particle cannot be a reconstruction-null object. Identity requires non-null standing-bearing persistence.
Bounded standing-localization	Particle identity must be localized rather than global, diffuse, or unbounded.
Continuation-neighborhood domains	Particle classes live across local continuation domains and admitted adjacency paths.
Quotient comparison carrier C_D	Particle identity must remain stable under admissible comparison and redescription.
Bounded perturbation transport	Particle persistence under displacement, disturbance, and restoration requires witness-mediated transport.
Field-carrier mediation Φ_D	Particle identity is a stable localized class within the carrier-mediated field role.
Boundary-trace fixation	Particle class witnesses must be boundary-compatible and trace-fixed, not default-selected.
Anchor-tensor-skin discipline	Particle labels and trajectory names are skin unless they carry standing-bearing role.
AMetric boundary discipline	Primitive object selectors, hidden cores, and default individuation are inadmissible.

The physics-facing implication is direct. In standard particle physics, particle names are tied to stable classification structures: mass class, charge, spin, gauge representation, flavor, lepton or baryon number, and other invariant witnesses, depending on context [16, 17, 14]. The analysis does not derive those specific witnesses. It derives why some stable witness of that general kind is required if a localized field-carrier configuration is to count as a particle at all.

3 Definitions

Definition 3.1 (Field-carrier assignment). A field-carrier assignment is the partial distributed carrier structure derived in the preceding paper,

$$\Phi_D : \text{Dom}(\Phi_D) \subseteq \text{LCD}_D \rightarrow \text{CarProf}_D.$$

For each admitted local continuation domain $U \in \text{Dom}(\Phi_D)$, $\Phi_D(U)$ is a carrier profile mediating localized persistence under bounded perturbation transport. The assignment is field-role structure, not field substance: admissible modifications of the carrier affect the standing-bearing transport organization of the regime.

Definition 3.2 (Localized carrier configuration). A localized carrier configuration over a local continuation domain U is a tuple

$$\kappa_U = (U, \Phi_D(U), \sigma_U, \chi_U, \beta_U).$$

Here U is the local continuation domain, $\Phi_D(U)$ is the field-carrier profile on that domain, σ_U is the standing signature retained by the configuration, χ_U is a boundary/classification witness by which the configuration remains distinguishable from other localized classes, and β_U is the bounded transport/restoration profile preserving the configuration across admissible continuation.

The tuple is not assembled ad hoc. Each component is inherited from a prior forced role:

Component	Prior forced role
U	continuation-neighborhood domains
$\Phi_D(U)$	field-carrier mediation
σ_U	standing preservation
χ_U	boundary/classification witness
β_U	bounded transport/restoration profile

Thus a localized carrier configuration is not a label attached to a domain. It is the local package of previously derived physical roles required for stable particle identity.

Definition 3.3 (Standing signature). The standing signature σ_U of κ_U is the standing-bearing identity profile retained by the configuration under admissible continuation. It is not a bare label. A string, index, coordinate, or name functions as a standing signature only if it tracks a standing-bearing role preserved by the regime.

Definition 3.4 (Boundary/classification witness). The boundary/classification witness χ_U is the admissible witness by which κ_U remains class-distinguishable under boundary comparison, quotient-preserving redescription, and admitted transport. Later physical specializations may realize this role by charge, spin, mass class, flavor, gauge representation, or other invariant classifications. None of those specific structures is assumed here.

Definition 3.5 (Transport/restoration profile). The transport/restoration profile β_U records the admitted perturbation-transport and restoration operations under which the localized configuration preserves its standing signature and boundary/classification witness. It is inherited from bounded perturbation transport, not added as an external trajectory rule.

Definition 3.6 (Particle-role equivalence). Let κ_U and κ_V be localized carrier configurations. Write

$$\kappa_U \equiv_{\Pi} \kappa_V$$

iff there is an admissible field-carrier transport/restoration chain from U to V that preserves

$$(\sigma, \chi, \beta, Q, I, \partial, T),$$

namely standing signature, boundary/classification witness, transport/restoration profile, quotient comparison class, identity-memory, boundary trace, and admitted transport role. The corresponding particle-role class is

$$\Pi_D = [\kappa_U]_{\equiv_{\Pi}}.$$

The relation $\equiv_{\Pi}$ is not reachability alone. Bounded transport may be directed, partial, or irreversible. Particle-role equivalence is defined by preservation of the same standing-bearing localized class under the admissible comparisons and continuations of the regime. This prevents particle identity from collapsing into mere motion along a path.

4 Failure of primitive identity candidates

The first task is negative. The usual intuitive sources of particle identity do not suffice once particle identity is required to persist through field-carrier mediation, bounded perturbation transport, and boundary-compatible comparison.

Lemma 4.1 (Primitive objecthood insufficiency). *A primitive object cannot supply particle identity unless it preserves standing signature, boundary/classification witness, quotient comparison, identity-memory, restoration profile, and transport role across admissible continuation. If it preserves those roles, it is no longer identity by primitive objecthood alone; it instantiates a particle-role class.*

Proof. Primitive objecthood posits an identity bearer before specifying the standing-bearing continuation conditions under which that identity remains evaluable. But a localized physical identity must survive admissible transport, perturbation, restoration, redescription, and boundary comparison. These require the preservation of the role components $(\sigma, \chi, \beta, Q, I, \partial, T)$. If the primitive object does not preserve them, it is not particle-identifiable in the realized regime. If it does preserve them, the identity is supplied by the preserved localized persistence class, not by primitive objecthood as an independent source. $\square$

Lemma 4.2 (Point-occupancy collapse). *A particle cannot be identified with mere occupancy of a point or local domain. Point occupancy does not by itself preserve class identity, boundary witness, restoration status, field-carrier mediation, or transport profile. If those roles are added, the result is a localized persistence class rather than primitive point occupancy.*

Proof. A point or local domain can locate a configuration, but location is not identity. Two configurations may occupy distinct domains while preserving the same particle-role class under admitted transport; conversely, a configuration may occupy a domain without preserving the standing signature or classification witness of a prior configuration. Point occupancy becomes identity-relevant only when it is coupled to preserved carrier-mediated roles. That coupling is precisely the particle-role equivalence structure. $\square$

Lemma 4.3 (Trajectory-label collapse). *A trajectory or worldline label can describe a continuation chain, but it cannot by itself carry particle identity. Without preserved carrier-mediated standing signature and boundary/classification witness, a trajectory label is skin.*

Proof. A trajectory label records an ordering or route of continuation. It does not determine what persists through the route. If the standing signature, classification witness, restoration profile, and quotient comparison class are not preserved, the label can connect unrelated carrier configurations. If they are preserved, the trajectory label is descriptive of an already established particle-role class. It is not the source of identity. $\square$

Lemma 4.4 (Field-label insufficiency). *A label attached to a field-carrier profile does not constitute particle identity unless the localized carrier configuration remains stable under admissible transport, restoration, boundary comparison, and quotient-preserving continuation.*

Proof. Field-carrier mediation supplies the distributed support required for localized persistence. It does not, by itself, individuate particle classes. A field label is merely notational unless it tracks a localized carrier configuration whose standing signature and boundary/classification witness remain stable under the admissible transformations of the regime. The label itself does not mediate

transport, restoration, standing preservation, quotient continuity, or persistence identity. It can classify a particle-role class only after those roles are already preserved. The particle role is therefore not a label on a field; it is a stable localized class within the field-carrier regime. $\square$

These lemmas jointly remove the main primitive identity candidates. Particle identity must be recovered from stability of localized carrier-mediated persistence, not posited as object substance, point occupancy, path syntax, or field notation.

5 Stable localization and particle-role equivalence

Proposition 5.1 (Stable localization requirement). *If a localized carrier-mediated configuration is particle-identifiable, then its standing signature and boundary/classification witness remain stable across admitted local continuation domains.*

$$\text{ParticleIdent}_D(\kappa_U) \Rightarrow \text{StableLoc}_D(\kappa_U).$$

Proof. A particle-identifiable configuration must remain the same physical class under admissible continuation. Same-class retention requires more than local support. It requires that the configuration's standing signature and boundary/classification witness remain jointly reconstructible after transport, restoration, and boundary comparison. If either component fails, the continuation does not preserve particle identity; it either produces an unclassified carrier configuration, resets identity, or imports an external selector. Hence particle identifiability implies stable localization of the preserved class. $\square$

Proposition 5.2 (Particle-role equivalence). *The relation $\equiv_\Pi$ is an admissible equivalence relation on localized carrier configurations whenever same-class preservation is defined by preservation of $(\sigma, \chi, \beta, Q, I, \partial, T)$ under admissible field-carrier transport/restoration chains.*

Proof. Reflexivity follows from identity continuation of a localized carrier configuration under the null admissible transport/restoration chain, preserving all listed role components. Symmetry is not assumed as reversal of directed transport. It follows from the definition of same preserved class: if κ_U and κ_V preserve the same standing signature and boundary/classification witness under admissible comparison, then each is a representative of the same class. Transitivity follows from admissible composition of transport/restoration chains when intermediate boundary traces, quotient comparison classes, and standing signatures remain stable. Therefore $\equiv_\Pi$ partitions the admissible localized carrier configurations into particle-role classes. $\square$

Remark 5.3 (Directed transport versus class equivalence). The proof does not require all transports to be reversible. Transport reach can be directed or partial. Equivalence is defined by preserved class under the admissible comparisons of the regime, not by bidirectional motion along every path. This is essential for compatibility with later irreversible, dissipative, or measurement-linked structures.

Proposition 5.4 (Persistence-class identity). *If $\kappa_U \equiv_\Pi \kappa_V$, then admissible modification of either representative changes the same standing-bearing transport/restoration organization of the regime, insofar as the modification is class-relevant. Hence $\Pi_D = [\kappa_U]_{\equiv_\Pi}$ is not merely a descriptive grouping; it is the persistence-bearing identity structure tracked by the representatives.*

Proof. By definition, $\kappa_U \equiv_{\Pi} \kappa_V$ only when the representatives preserve $(\sigma, \chi, \beta, Q, I, \partial, T)$ under admitted field-carrier transport/restoration chains. A class-relevant modification of one representative changes one or more of these preserved roles. Since the other representative is equivalent only by preserving the same roles, that modification changes the admissible transport/restoration organization of the class rather than merely changing a label. A purely descriptive class could be relabeled without altering persistence. A particle-role class cannot: its representatives stand or fail together with respect to the preserved identity roles. Therefore the equivalence class is physically operative as persistence-class identity. $\square$

6 Particle persistence-class induction

Theorem 6.1 (Particle Persistence-Class Theorem). *In a fixed admissible realized persistence regime with field-carrier mediation, bounded perturbation transport, local continuation domains, and quotient comparison carrier, any localized carrier-mediated configuration that remains standing-identifiable under admissible continuation induces a particle-role class. The class is induced by the impossibility of transport-stable particle identity without preservation of the carrier-mediated roles listed above:*

$$(\Phi_D, \text{BoundTrans}_D, C_D, \text{LCD}_D) \wedge \text{StableLoc}_D \wedge \text{IdRet}_D \Rightarrow \exists \Pi_D.$$

Equivalently,

$$\text{ParticleRole}_D \equiv \Pi_D = [\kappa_U]_{\equiv_{\Pi}}.$$

Proof. By the field-carrier result, localized persistence under bounded perturbation transport is mediated by a distributed carrier assignment Φ_D . A localized carrier configuration $\kappa_U = (U, \Phi_D(U), \sigma_U, \chi_U, \beta_U)$ therefore supplies local support, carrier mediation, standing signature, boundary/classification witness, and transport/restoration profile. By the failure lemmas, primitive objecthood, point occupancy, trajectory labels, and field labels are insufficient to ground particle identity by themselves. By the stable localization requirement, a particle-identifiable configuration must preserve standing signature and boundary/classification witness across admitted continuation. By particle-role equivalence, configurations preserving $(\sigma, \chi, \beta, Q, I, \partial, T)$ under admissible field-carrier transport/restoration chains belong to the same class. By the persistence-class identity proposition, this class is not merely a bookkeeping partition: class-relevant modification changes the same standing-bearing transport/restoration organization tracked by the representatives. Hence the admissible configurations induce stable localized persistence classes $\Pi_D = [\kappa_U]_{\equiv_{\Pi}}$. This class is the minimal particle role. $\square$

The theorem identifies the object role without postulating a primitive object. The localized persistence class is not a descriptive substitute for particle identity; it is the condition under which particle identity becomes physically meaningful in a realized regime.

7 Localized persistence class versus object substance

A common order of explanation treats a particle as a primitive object that persists, moves, interacts, and is later described by fields or quantum states. The present order is different. Persistence structure comes first. Field-carrier mediation supplies the support for localized persistence under bounded transport. A particle is then the stable localized equivalence class of carrier-mediated configurations. Objecthood is not denied; it is reconstructed as the stabilized appearance of a persistence class.

This relocation is physics-facing. In classical mechanics, the object picture can be effective because trajectories and material localization often track the same persistence class. In relativistic quantum field theory, particle descriptions are already more delicate: particle number, localization, detector outcomes, asymptotic states, and field excitations do not reduce to a single primitive object picture [15, 16, 17]. The present analysis derives the more primitive role common to such uses: a particle must be a localized class whose identity is stable under the transformations that count as admissible in the regime. This is not a descriptive class that merely groups already given objects; it is a persistence class whose role is to preserve the object-identity conditions themselves.

8 Relation to field-carrier modes

A particle-role class is supported by the field-carrier regime. It is natural to call such a class a localized persistence mode of Φ_D , provided that “mode” is understood structurally rather than as a specific harmonic, Fock-space, or eigenmode construction. Formally, Π_D is represented by localized carrier configurations κ_U , each of which includes $\Phi_D(U)$. Thus particle identity depends on field-carrier mediation, but is not identical to the entire field assignment.

The distinction is important. A field-carrier assignment can support many localized configurations, overlapping carrier structures, non-particle background roles, or field regimes with no stable localized particle class. Conversely, a particle-role class can persist through changes in local representative κ_U so long as the standing signature and boundary/classification witness remain preserved. Particle identity is therefore not “the field” and not “a label on the field.” It is a stable localized persistence class within the field-carrier structure.

9 Relation to perturbation transport

Particle persistence across motion, disturbance, decay-like restructuring, or restoration cannot be understood without bounded perturbation transport. If $\kappa_U \equiv_{\Pi} \kappa_V$, the relevant admissible chain must preserve the same particle-role class under field-carrier transport. In particular, for admitted transport witnesses $\tau_D(U, V, p)$, the source and target configurations must preserve the class data

$$(\sigma, \chi, \beta, Q, I, \partial, T).$$

A path can carry a localized configuration only if the carrier profiles mediate the standing signature and boundary/classification witness along the path. Otherwise the path is merely descriptive, and the alleged particle is replaced by unrelated local carrier data.

This gives the particle role a controlled relation to future dynamics. The analysis does not derive equations of motion or scattering laws. It fixes the identity condition that any later dynamics must preserve or lawfully transform.

10 Boundary/classification witnesses and future quantum numbers

The abstract witness χ_U is the bridge from the present theorem to particle classification in later physics. In standard applications, particle identities are stabilized by classification data such as electric charge, color representation, weak isospin, spin, mass class, flavor, lepton number, baryon number, helicity sector, or other invariant roles, depending on the regime. The present theorem

does not derive those specific structures. It derives why some stable boundary/classification witness is required.

This distinction preserves the force of the physical claim while fixing its scope. The result has a direct particle-physics reading: it identifies the role that real particle classifiers must play: they are admissible witnesses by which localized carrier configurations remain class-identifiable under transport, redescription, restoration, interaction, and boundary comparison. Later Standard Model specialization can then ask which witnesses are forced by gauge structure, representation constraints, mass/inertia structure, and quantization.

11 No hidden-variable particle

Corollary 11.1 (No Hidden-Variable Particle). *A particle-role class is not a Bell-local hidden variable, deterministic local state, subquantum bead, or outcome-determining object. It is a stable localized persistence class supported by field-carrier mediation and boundary-compatible continuation.*

Proof. The particle-role class is defined by admissible preservation of class data under field-carrier transport and boundary comparison. This does not assert factorization of quantum probabilities, local deterministic outcome control, or separability of all tensor-level compatibility relations. Nonclassical boundary compatibility, entanglement, and measurement-context dependence are not excluded by the existence of stable localized persistence classes. The class supplies particle identity, not a hidden-variable ontology. $\square$

This qualification fixes the result's relation to quantum locality. Deriving particle identity does not return the theory to naive corpuscular mechanics. It derives the minimal identity role that later quantum particle descriptions, including nonclassical and entangled descriptions, must still respect when they speak of identifiable particle classes.

12 Relation to measurement records

Measurement records are downstream fixations of particle-role events. The present result does not deny records, detection events, measurements, or empirical particle observations. A detector record may fix that a configuration satisfying a particle-role class occurred, transitioned, interacted, or was absorbed. The claim is only that the record is not the primitive source of particle identity. A record-only account cannot explain what the record is a record of unless the class-relative identity conditions are already admissible. Conversely, a particle-role class need not be interpreted as a definite measurement outcome in every regime. It is the persistence-class identity that makes such outcomes meaningful when record construction is available.

13 Relation to Standard Model particles

Standard Model particles add much more structure than the present theorem derives. They involve gauge representations, charge assignments, spin, chirality, mass generation, flavor structure, quantum statistics, field quantization, interactions, and symmetry-breaking regimes [18, 19, 20, 16, 17]. The present result does not derive those features. It derives the role they must occupy: stable localized particle identity requires a class witness preserved across admissible field-carrier transport and boundary comparison.

This role is recognizable in ordinary particle physics. A particle description must be locally supportable, transport-stable, identity-preserving, boundary-distinguishable, and compatible with the field structures that mediate its persistence. Standard quantum and Standard Model formalisms supply specialized machinery for those roles; they do not eliminate the need for the roles.

In that sense the theorem is positive rather than merely eliminative. Standard field and particle formalisms make particle descriptions locally supportable, transport-stable, identity-preserving, boundary-distinguishable, and compatible with the carrier structures that mediate their persistence. The present analysis derives the need for those roles before specifying the particular representation-theoretic and dynamical machinery that realizes them.

Thus the theorem is not a replacement for Standard Model classification. It is a prior condition on what Standard Model labels are doing. A label such as “electron” is not merely a word, nor merely point occupancy, nor merely a detector event. It names a stable particle-role class whose detailed witnesses are supplied by the deeper field, gauge, mass, and quantum structures. Later papers in the arc can therefore ask how gauge boundary fixation, mass/inertia compression, and vacuum structure further specialize χ_U , β_U , and σ_U .

14 Multiplicity and individuation

Particle identity and particle multiplicity are distinct questions. A particle-role class Π_D identifies what it is for representatives to count as the same particle role. Multiplicity asks when two localized representatives instantiate distinct tokens of a class rather than two descriptions of one token. The analysis does not derive quantum statistics, symmetrization, antisymmetrization, or occupation-number structure. It does identify the prior requirement: multiplicity must be grounded in admissible separability of localized carrier configurations, not in primitive object counting.

If two localized configurations share the same class-defining witness χ but remain separable under continuation-neighborhood domains, bounded transport profiles, and carrier-mediated identity-memory, then the regime may support multiple instances of the same particle role. If the configurations collapse under admissible comparison into a single equivalence representative, then counting them as two particles is a bookkeeping artifact. If they differ in standing signature or classification witness, they belong to distinct particle-role classes. Thus individuation is neither primitive numerical tagging nor mere spatial separation. It is stable class-relative distinguishability under the already-derived local-domain, carrier, and transport structure.

This distinction controls later quantum specialization. Standard quantum theory treats particle identity and multiplicity through state-space, statistics, and field-excitation structures [14, 16]. The present analysis does not reproduce that machinery. It supplies the role condition those formalisms must respect: particle counting must be compatible with stable localized persistence classes and not merely with arbitrary labels.

15 Interaction and class preservation

A particle-role class need not remain isolated. In a realized physical regime, localized configurations may undergo admitted perturbation transport, boundary exchange, restoration, or interaction-like coupling. The class persists only when the class-defining tuple is preserved through those processes. If an interaction changes the standing signature, boundary/classification witness, restoration profile, or quotient comparison in a class-breaking way, then the original particle role terminates

or transforms into another role. If those invariants are preserved, the same particle class persists through the interaction.

This gives a structural distinction between change and loss of identity. Motion, deformation, local perturbation, or carrier-mediated transport need not destroy particle identity. They are class-preserving when they remain inside $\equiv_{\Pi}$. By contrast, identity reset, hidden transfer, cross-domain import, or boundary-witness failure does destroy the identity claim. This is the particle-level analogue of the earlier distinction between admissible transport and pathless global effect.

The analysis does not derive scattering theory. It identifies the pre-scattering identity condition: before amplitudes, cross sections, reaction channels, or detector records can be assigned, the regime must already distinguish class-preserving continuation from class-changing transformation.

16 Particle class and excitation language

The common statement that particles are excitations of fields is not assumed here. It becomes intelligible only after field-carrier mediation has been derived. Within the present framework, a particle-role class is a stable localized class of field-carrier configurations. In later specializations, such classes may be represented as excitations, modes, quanta, poles, irreducible representation types, or asymptotic states. Those are formal enrichments of the class role, not its source.

This order matters. If one begins with excitation language, the field structure and state-space machinery are already in place. The present analysis instead identifies the role a particle must occupy before those formalisms are introduced. The answer is less specialized and more basic: a particle is a localized persistence class of carrier-mediated configurations preserving class identity under admissible continuation. Excitation formalisms may later model such classes, but the persistence-class role is already required.

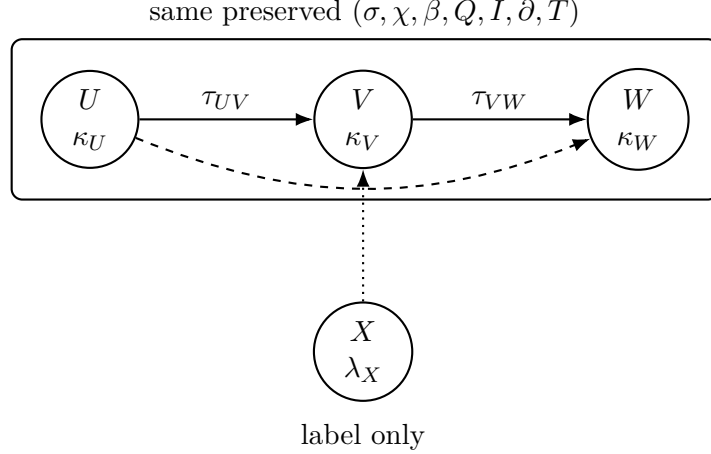
17 Finite witness model

The finite model below is a witness model only. It instantiates one admissible realization of persistence-class mediation and does not define the general particle-role structure or identify particles with graph vertices.

Let

$$G_D = (V, E, \Phi, \kappa, \equiv_{\Pi}),$$

where V is a finite family of local continuation domains, E is a set of admitted carrier-transport links, $\Phi(v)$ is the field-carrier profile at v , κ_v is a localized carrier configuration, and $\equiv_{\Pi}$ identifies configurations preserving the same particle-role data.



The solid path represents admitted carrier transport preserving the class data. The dashed link is admissible only if it preserves the same particle-role data as the composed path. The dotted label-only connection from X does not establish particle identity because no carrier-mediated class preservation is supplied.

The collapse cases are immediate:

- (i) a primitive object without a witness gives unsupported identity;
- (ii) point occupancy alone gives local presence without transport-stable class;
- (iii) a trajectory label alone is skin;
- (iv) a field label alone is bookkeeping;
- (v) a hidden core imports an inaccessible reservoir;
- (vi) a measurement record alone fixes a downstream event, not the identity source;
- (vii) stable localized carrier configurations preserving $(\sigma, \chi, \beta, Q, I, \partial, T)$ yield a particle-role class.

18 Route exhaustion

Theorem 18.1 (Particle Identity Route Exhaustion). *Any candidate source of particle identity in a fixed admissible realized persistence regime with field-carrier mediation reduces to one of the following broad classes: persistence-class identity, primitive-object identity, label-only identity, hidden-core identity, record-only identity, transport-free identity, or cross-domain imported identity. All except persistence-class identity fail to provide admissible particle identity.*

Proof. Primitive-object identity fails by the primitive objecthood insufficiency lemma; point occupancy is a special case in which local presence is mistaken for identity. Label-only identity includes trajectory, worldline, and field labels; these fail because path syntax and field names do not carry standing-bearing identity unless they track a stable localized carrier configuration, in which case they reduce to persistence-class identity. Hidden-core identity violates admissible boundary-trace and AMetric discipline unless the core becomes accessible, in which case it becomes part of the class witness. Record-only identity fails because measurement records are downstream fixations and cannot be the source of identity. Transport-free identity fails because a particle class

must remain identifiable under bounded perturbation transport and restoration. Cross-domain imported identity exits fixed-domain admissibility. The remaining candidate is a stable localized persistence class $\Pi_D = [\kappa_U]_{\equiv \Pi}$, which is admissible by the particle persistence-class theorem. $\square$

19 Normal form

The result can be summarized as

$$\boxed{(\Phi_D, \text{BoundTrans}_D, C_D, \text{LCD}_D) \wedge \text{StableLoc}_D \wedge \text{IdRet}_D \Rightarrow \text{ParticleRole}_D}$$

with

$$\text{ParticleRole}_D \equiv \Pi_D = [\kappa_U]_{\equiv \Pi}, \quad \kappa_U = (U, \Phi_D(U), \sigma_U, \chi_U, \beta_U).$$

The physical role is that particle identity first enters realized physics as stable localized persistence-class identity within the field-carrier regime. Primitive objecthood, point occupancy, worldline labels, field labels, hidden cores, and measurement records may describe or fix particle roles only after the persistence-class identity conditions are already available.

20 Conclusion

The preceding realized-physics papers established the non-null floor of physical realization, the origin of physical locality as bounded standing-localization, the origin of local-domain organization as continuation-neighborhood structure, the metric role as quotient-stable comparison, the finite-propagation constraint role as bounded perturbation transport, and the field role as distributed carrier mediation. The present paper isolates the minimal persistence-class role required for stable localized identity under field-carrier mediation and bounded perturbation transport.

The result is physics-facing but role-first. Physical particles are later specializations of $\Pi_D = [\kappa_U]_{\equiv \Pi}$, enriched by gauge representation, spin, mass, field dynamics, quantum statistics, measurement records, and interaction structure. Those enrichments are not assumed here. What is derived is the object-role condition they must satisfy: stable localized carrier-mediated persistence class identity.

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